Newton's Laws

Section 1: Newton's Second Law ($F = ma$)

1 A constant resultant force of 36 N acts on a mass of 4 kg. What is the acceleration produced?

a. $9\ \text{m/s}^2$

b. $144\ \text{m/s}^2$

c. $4\ \text{m/s}^2$

d. $6\ \text{m/s}^2$

2 A car of mass 1200 kg is traveling at 25 m/s when the brakes are applied, bringing it to rest in 50 m. Find the magnitude of the braking force.

a. 7500 N

b. 6000 N

c. 9000 N

d. 15000 N

3 An elevator of mass 800 kg accelerates upwards at $1.5\ \text{m/s}^2$. Taking $g = 10\ \text{m/s}^2$, find the tension in the cable.

a. 9200 N

b. 8000 N

c. 6800 N

d. 1200 N

Section 2: Connected Particles over a Pulley

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4 Two masses of 3 kg and 2 kg are connected by a light inextensible string passing over a smooth fixed pulley (Atwood machine). Taking $g = 10\ \text{m/s}^2$, find the acceleration of the system.

a. $2\ \text{m/s}^2$

b. $1\ \text{m/s}^2$

c. $4\ \text{m/s}^2$

d. $5\ \text{m/s}^2$

5 For the system in the previous question, find the tension T in the string.

a. $24\ \text{N}$

b. $25\ \text{N}$

c. $20\ \text{N}$

d. $30\ \text{N}$

6 What is the total force exerted by the string on the pulley in the previous system?

a. $48\ \text{N}$ downwards

b. $24\ \text{N}$ downwards

c. $50\ \text{N}$ downwards

d. $40\ \text{N}$ downwards

Section 3: Motion on Rough Inclined Planes

7 A block of mass 5 kg is pulled up a smooth slope inclined at 30° to the horizontal by a force of 40 N acting parallel to the slope. Taking $g = 10\ \text{m/s}^2$, find the acceleration of the block.

a. $3\ \text{m/s}^2$

b. $5\ \text{m/s}^2$

c. $2\ \text{m/s}^2$

d. $4\ \text{m/s}^2$

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8 If the slope in the previous question is rough with $\mu = 0.2$, what is the new acceleration of the block?

a. $1.27\ \text{m/s}^2$

b. $3.00\ \text{m/s}^2$

c. $2.15\ \text{m/s}^2$

d. $0.85\ \text{m/s}^2$

9 A sled of mass 20 kg slides down an incline of 25° at constant speed. What is the coefficient of kinetic friction μ?

a. $\mu = \tan(25^\circ) \approx 0.466$

b. 0.423

c. 0.906

d. 0.500

Section 4: Connected Systems with Pulleys & Tables

10 A block A of mass 4 kg rests on a smooth horizontal table and is connected by a light string over a smooth pulley at the edge of the table to a hanging block B of mass 1 kg. Taking $g = 10\ \text{m/s}^2$, find the acceleration.

a. $2\ \text{m/s}^2$

b. $2.5\ \text{m/s}^2$

c. $1.5\ \text{m/s}^2$

d. $10\ \text{m/s}^2$

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11 If the table in the previous system has coefficient of friction $\mu = 0.15$, find the acceleration of the system.

a. $0.8\ \text{m/s}^2$

b. $1.2\ \text{m/s}^2$

c. $1.5\ \text{m/s}^2$

d. $0.5\ \text{m/s}^2$

12 A car of mass 1000 kg tows a trailer of mass 500 kg with acceleration $1.2\ \text{m/s}^2$. Resistance to motion is 300 N on the car and 200 N on the trailer. Find the tension in the towbar.

a. 800 N

b. 600 N

c. 1100 N

d. 1500 N

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